Transport processes

Down a gradient

1 Diffusion

Net movement: gas/dissolved particles, high → low concentration, down the gradient.

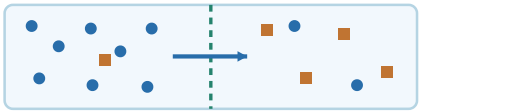
Random movement; **no respiration energy.** Examples: O_2, CO_2; urea from cells into blood plasma. At equilibrium particles move, but no net movement.

2 Faster diffusion

Steeper gradient: greater net movement/second. **Hotter:** more kinetic energy.

Larger area: more particles cross. **Thinner barrier:** shorter diffusion path.

3 Osmosis

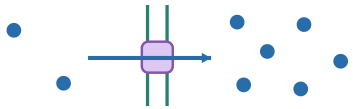


Water: dilute → concentrated solution through a **partially permeable membrane.**

Dilute side has more water relative to solute. Water crosses; some solutes cannot. No respiration energy. Movement is both ways; arrows show the net direction.

Energy & cell changes

4 Active transport & its uses



Low → high; respiration energy

● solute | membrane ▣ protein

Substances move **low → high,** against the gradient, via membrane proteins.

Needs **energy from respiration,** like moving uphill. Diffusion/osmosis move down gradients; only osmosis is water-only.

Roots: take up minerals for growth even when soil concentration is lower.

Gut: absorbs sugars into blood even when blood concentration is higher; sugars fuel respiration. Many mitochondria support high energy demand.

5 Water gain or loss

Dilute outside → water enters. Plant: vacuole fills, **turgid;** wall prevents bursting. Animal: may swell/burst.

Concentrated outside → water leaves. Plant: **flaccid;** severe loss → plasmolysis (membrane pulls from wall). Animal: shrinks.

Investigate osmosis

6 Osmosis practical

Vary salt/sugar concentration; use equal-size potato cylinders. Weigh initially; equal solution volumes, including distilled water.

Control tissue source, size, time and temperature. **Blot consistently,** reweigh; repeat and calculate means. Blotting removes surface liquid; cut safely.

7 Calculate & interpret

% change = (final − initial) ÷ initial × 100. 2.50 → 2.20 g: −0.30 ÷ 2.50 × 100 = **−12%.**

Gain +; loss −. Plot concentration (x), mean % change (y). Best-fit curve's 0% crossing = **no net movement,** not no movement.

Rate = change ÷ time: 0.12 g ÷ 30 min = **0.004 g/min.** Check anomalies; repeat/test nearer zero.